Pointlike Scattering from Fermionic Open Strings and a Bootstrap Interpretation of Quarks

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A finite open string can scatter, at tree level, more pointlike than a generic composite fermion. For a charged Ramond state we show that the Møller-like amplitude is the massless gauge-theory amplitude multiplied by a Veneziano factor. The first string correction is $O(\alpha'^2 E^4)$, with no dimension-six contact term in the same-brane sector. We derive the angular signal and discuss the limited but concrete sense in which this realizes a quark–diquark bootstrap.

Motivation.— The original dual-resonance program was simultaneously a theory of extended hadrons and an S-matrix bootstrap: poles in one channel were to be understood as bound states in a crossed channel. Ramond's fermionic extension placed spin-1/2 states on linear trajectories [1], while the Neveu–Schwarz (NS) model supplied the bosonic sector [2]. Aharonov, Casher, and Susskind subsequently obtained the NS and fermionic sectors from even and odd numbers of spin-1/2 partons and explicitly computed a meson–baryon amplitude [3]. Strings with quark quantum numbers attached to their ends, including spin-1/2 endpoint models, were constructed by Bars and Hanson [4]. Independently, Miyazawa's meson–baryon supersymmetry [5] and its quark–diquark realization by Catto and Gürsey [6] exploit the elementary color fact

$$3_c \otimes 3_c = \bar{3}_c \oplus 6_c, \qquad \bar{3}_c \otimes \bar{3}_c = 3_c \oplus \bar{6}_c. \tag{1}$$

Thus an antisymmetric anti-diquark carries the color representation of a quark.

A sharper question can be asked without assuming a hadronic model: *how does a genuinely extended fermionic string reveal itself in elastic scattering?* A generic composite fermion has a low-momentum form factor $F(q^2) = 1 - q^2 r^2/6 + \cdots$, suggesting an $O(E^2/\Lambda^2)$ deviation. By contrast, the simplest supersymmetric open-string amplitude has no $O(\alpha')$ correction. Below we make this statement explicit by comparing four Ramond strings with massless Møller scattering and then state precisely what can, and cannot, be imported into a quark bootstrap.

Point-fermion benchmark.— For $e^-(p_1)e^-(p_2) \to e^-(p_3)e^-(p_4)$ and $m_e = 0$, define

$$s = (p_1+p_2)^2, \qquad t = (p_1-p_3)^2, \qquad u = (p_1-p_4)^2, \qquad s+t+u = 0. \tag{2}$$

The t- and u-channel photon graphs in Fig. 1(a) give

$$\mathcal{M}_{\rm QED} = e^2\left(\frac{J_t}{t} - \frac{J_u}{u}\right), \tag{3}$$

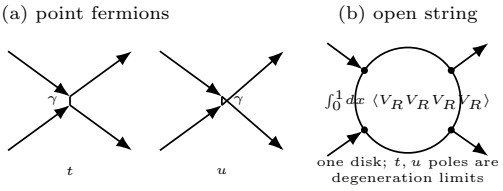


FIG. 1. (a) The two Møller graphs. (b) Four Ramond vertex operators on a disk. Point-particle propagator graphs arise at boundaries of the disk modulus rather than as distinct fundamental string diagrams.

where

$$J_t = (\bar{u}_3\gamma^\mu u_1)(\bar{u}_4\gamma_\mu u_2), \qquad J_u = (\bar{u}_4\gamma^\mu u_1)(\bar{u}_3\gamma_\mu u_2). \tag{4}$$

The minus sign is the exchange sign of the identical final fermions. Standard traces give

$$\overline{|\mathcal{M}_{\rm QED}|^2} = 2e^4\left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{2s^2}{tu}\right]. \tag{5}$$

With $t = -s(1-\cos\theta)/2$ and $u = -s(1+\cos\theta)/2$, including the $1/2!$ for identical particles,

$$\frac{d\sigma_{\rm QED}}{d\Omega} = \frac{\alpha^2}{2s}\,\frac{(3+\cos^2\theta)^2}{\sin^4\theta}. \tag{6}$$

Four Ramond strings.— For an open RNS string the massless Ramond vertex in the $-1/2$ picture is, suppressing compactification-dependent internal operators,

$$V_R^{(-1/2)}(u,k;x) = g_o\,\lambda\,u^A S_A(x) e^{-\phi(x)/2} e^{ik\cdot X(x)}, \tag{7}$$

with Chan–Paton matrix λ. Four such vertices supply the required disk superghost charge -2. After fixing three boundary positions by $PSL(2,\mathbb{R})$, the remaining modulus $x \in (0,1)$ generates an Euler beta integral. The standard result for any fixed cyclic ordering of massless open-superstring states is [7]

$$\mathcal{A}_{\rm str}(1,2,3,4) = \mathcal{A}_{\rm SYM}(1,2,3,4)\,V(s_{12},s_{23}), \tag{8}$$

where

$$V(a,b) = \frac{\Gamma(1-\alpha' a)\Gamma(1-\alpha' b)}{\Gamma[1-\alpha'(a+b)]}. \tag{9}$$

Equation (8) is the operative statement of "pointlike below the string scale": the complete spinor numerator and massless gauge poles are those of super-Yang–Mills; the extended object multiplies them by an analytic function containing the massive Regge completion. It is useful to see where the Gamma functions enter. For a cyclic ordering with adjacent invariants a, b, the Koba–Nielsen part of the gauge-fixed disk integral contains

$$I(a,b) = \int_0^1 dx\, x^{-\alpha' a-1}(1-x)^{-\alpha' b-1} = B(-\alpha' a, -\alpha' b). \tag{10}$$

Using $a+b+c=0$ and $\Gamma(1-z) = -z\Gamma(-z)$,

$$V(a,b) = \frac{\alpha' ab}{-c}\, B(-\alpha' a, -\alpha' b), \tag{11}$$

so the factors of a, b, c supplied by the spin-field kinematic correlator convert the beta-function poles into the familiar massless gauge poles plus a tower of massive poles. Near a crossed-channel pole $a = nM_s^2$ $(n = 1, 2, \ldots)$,

$$V(a,b) \underset{a\to nM_s^2}{\sim} \frac{1}{\alpha'(n-1)!}\, \frac{\prod_{k=0}^{n-1}(\alpha' b + k)}{a - nM_s^2}, \tag{12}$$

up to the convention-dependent overall sign inherited from the definition of Mandelstam variables. The residue is a degree-n polynomial, the standard Regge factorization into the spins carried at level n. At $a = 0$ the pole instead resides in $\mathcal{A}_{\rm SYM}$; $V(0,b) = 1$, so the massless residue is exactly the field-theory one.

To obtain a Møller-like charged state with the minimum amount of structure, take a coincident $U(2)$ stack and use the off-diagonal Ramond Chan–Paton matrices

$$\lambda_+ = E_{12}, \qquad \lambda_- = E_{21}, \qquad [Q, \lambda_\pm] = \pm\lambda_\pm, \tag{13}$$

for $Q = \mathrm{diag}(1,-1)/2$. This is a gauge-theory analogue of a charged fermion, not a realistic electron: at coincident branes it belongs to the non-Abelian adjoint. For the all-incoming crossing of same-sign scattering choose $\lambda_1 = \lambda_2 = \lambda_+$ and $\lambda_3 = \lambda_4 = \lambda_-$. Since $E_{12}^2 = E_{21}^2 = 0$, cyclic traces with equal charges adjacent vanish, while alternating traces survive. Thus the s-channel color ordering is removed, and the two nonzero orderings are $(1,3,2,4)$ and $(1,4,2,3)$. Their adjacent invariants are (t,u) and (u,t); because V is symmetric, the same factor multiplies both exchange terms. Consequently

$$\boxed{\mathcal{M}_{RR} = V(t,u)\, \mathcal{M}_{\rm gauge}} \tag{14}$$

with $\mathcal{M}_{\rm gauge}$ having exactly the current-current form in Eq. (3), up to the expected group normalization. This is the precise comparison with QED. The intermediate factorization states of $RR \to RR$ are NS states: at low energy the massless gauge boson, and above it the bosonic open-string tower. The fermionic external string therefore does not make single-fermion exchange appear in a diagonal fermion–fermion force; the world-sheet rule is the same even-fermion rule as in local field theory.

Observable string form factor.— In the physical Møller region $s > 0$, $t, u < 0$. Writing

$$z \equiv \alpha' s = \frac{s}{M_s^2}, \qquad c \equiv \cos\theta, \tag{15}$$

Eq. (9) becomes

$$V(z,c) = \frac{\Gamma[1+\frac{z}{2}(1-c)]\,\Gamma[1+\frac{z}{2}(1+c)]}{\Gamma(1+z)}. \tag{16}$$

Hence the exact tree-level prediction of the toy model is

$$\boxed{\frac{d\sigma_{RR}}{d\Omega} = \frac{d\sigma_{\rm gauge}}{d\Omega}\, |V(z,c)|^2.} \tag{17}$$

There are no physical Regge poles in this same-sign channel because $t, u < 0$ and the s-channel ordering is absent; the signal is a smooth angular deformation. The Regge poles become physical in crossed channels that admit s-channel string production.

The low-energy expansion follows from $\log\Gamma(1-x) = \gamma x + \sum_{n\geq 2}\zeta(n)x^n/n$:

$$V(t,u) = 1 - \zeta(2)\alpha'^2 tu + \zeta(3)\alpha'^3 stu + O(\alpha'^4), \tag{18}$$

$$\frac{d\sigma_{RR}}{d\sigma_{\rm gauge}} = 1 - \frac{\pi^2}{12}\left(\frac{s}{M_s^2}\right)^2 \sin^2\theta + O\left(\frac{s^3}{M_s^6}\right). \tag{19}$$

The absence of an $O(\alpha' E^2)$ term is the central experimental fact. In effective-field-theory language the leading same-stack oscillator correction is dimension eight, schematically

$$\Delta\mathcal{L}_8 \sim \frac{g^2}{M_s^4}(\bar\psi\gamma^\mu\partial^\nu\psi)(\bar\psi\gamma_\mu\partial_\nu\psi) + \cdots, \tag{20}$$

not a generic dimension-six four-fermion contact term. This distinction is known in D-brane phenomenology: fermions localized at different brane intersections can instead generate dimension-six operators, whereas same-stack open-string matter has the dimension-eight oscillator behavior used here [9].

Equation (19) converts immediately into an experimental scale. If a central-angle measurement at center-of-mass energy $E_{\rm cm} = \sqrt{s}$ agrees with the point-fermion result to fractional accuracy ϵ, then

$$M_s \gtrsim E_{\rm cm}\left(\frac{\pi^2}{12\epsilon}\right)^{1/4} \qquad (\theta = \pi/2). \tag{21}$$

Thus 1% sensitivity probes only $M_s \simeq 3.0 E_{\rm cm}$, while 0.1% gives $5.35 E_{\rm cm}$: energy is far more valuable than precision

because the reach grows only as $\epsilon^{-1/4}$. Conversely, the angular structure is diagnostic: the correction vanishes in the forward limit and is maximal at 90°. At that angle the exact form factor is especially simple,

$$V_{90}(z) = \frac{\Gamma(1+z/2)^2}{\Gamma(1+z)}, \tag{22}$$

and illustrates how rapidly the asymptotic string regime turns on:

$z = s/M_s^2$	0.10	0.25	0.50	1.0	2.0
$\lvert V_{90}\rvert^2$	0.992	0.957	0.859	0.617	0.250

This pattern is not equivalent to the usual compositeness ansatz. A dimension-six current-current interaction $\eta(\bar\psi\gamma_\mu\psi)^2/\Lambda^2$ interferes with gauge exchange at relative order s/Λ^2 and is the basis of many quoted "quark radius" or compositeness limits. The present disk amplitude instead starts at s^2/M_s^4. Current collider limits on conventional dimension-six quark and lepton contact operators extend to scales of order tens of TeV in several channels [11], but they therefore cannot be translated by the replacement $\Lambda \to M_s$; the angular and energy dependence must be fitted to Eq. (17) or to its dimension-eight expansion.

At fixed angle and $z \gg 1$, Stirling's formula gives, with $x = (1-c)/2$,

$$\begin{aligned} V &\sim \sqrt{2\pi z x(1-x)}\, e^{-zH(x)}, \\ H(x) &= -x\ln x - (1-x)\ln(1-x) > 0, \end{aligned} \tag{23}$$

so the extended nature ultimately appears as the familiar exponential softness of hard string scattering. At 90°, $V \sim \sqrt{\pi z/2}\, 2^{-z}$.

A caution about terminology is important here. Equation (17) is an on-shell four-point statement, not a direct computation of a static charge radius. In ordinary compositeness analyses the latter is encoded through off-shell current form factors or, at energies above bound-state scales, through local four-fermion operators. What is unusually soft in the present string sector is specifically the tree-level disk correction to massless scattering: supersymmetry and the open-string amplitude remove the dimension-six oscillator term. Compactification effects, Kaluza–Klein exchange, brane intersections, or loops need not share that cancellation. Thus "pointlike through E^2/M_s^2" is a precise amplitude statement rather than the claim that a fundamental string possesses zero spatial extent.

Where a fermionic mediator does appear.— The elastic $RR \to RR$ process above also clarifies the question of a fermion mediating a force. A local Lorentz-invariant vertex is Grassmann even; the open-string counterpart is that the elementary disk three-point couplings are of NS–NS–NS and NS–R–R type. Hence four external Ramond states factorize through NS states. A single Ramond state can nevertheless be exchanged in a mixed process

$$R_1 + NS_2 \longrightarrow NS_1 + R_2, \tag{24}$$

for which each end of the propagator converts a bosonic state into a fermionic one. The field-theory limit makes the mechanism transparent. Introduce a heavy fermion χ and interactions

$$\mathcal{L}_{\rm int} = g_1\phi_1^\dagger\bar\chi\psi_1 + g_2\phi_2^\dagger\bar\chi\psi_2 + \text{H.c.} \tag{25}$$

Then the crossed conversion amplitude contains

$$\mathcal{M}_R = g_1 g_2^*\,\bar u_2 \frac{q+M}{q^2-M^2} u_1. \tag{26}$$

For $|q^2| \ll M^2$,

$$\mathcal{M}_R = -\frac{g_1 g_2^*}{M}\bar u_2 u_1 - \frac{g_1 g_2^*}{M^2}\bar u_2 q\, u_1 + \cdots, \tag{27}$$

which is reproduced by the local operator

$$\mathcal{L}_{\rm eff} = \frac{g_1 g_2^*}{M}\phi_1^\dagger\phi_2\,\bar\psi_2\psi_1 + \text{H.c.} + \cdots. \tag{28}$$

Thus the zero-range limit does not forbid fermionic mediation; it makes it a contact conversion operator. What remains forbidden is a single fermion generating a diagonal static potential between unchanged ordinary fermionic sources. In the string language the distinction is topological: an R strip can connect mixed NS/R boundaries, while the four-R disk degenerates through an NS strip. This is exactly the channel distinction obscured by the slogan that "forces are mediated by bosons."

Crossing and realistic color.— The particularly transparent factorization in Eq. (14) relies on the alternating $U(2)$ Chan–Paton assignment, for which both surviving orderings carry the identical function $V(t,u)$. It should not be mistaken for a theorem that every four-quark observable is obtained by multiplying the full QCD answer by one scalar form factor. In a general non-Abelian process the full disk amplitude is a sum

$$\mathcal{A}_4 = g_o^2 \sum_{\sigma/\mathbb{Z}_4} \mathrm{Tr}(\lambda_{\sigma_1}\lambda_{\sigma_2}\lambda_{\sigma_3}\lambda_{\sigma_4}) A_{\rm YM}(\sigma)\, V_\sigma, \tag{29}$$

with $V_\sigma = V(s_{\sigma_1\sigma_2}, s_{\sigma_2\sigma_3})$. Different color orderings can therefore carry different functions before the color sum. Nevertheless every $V_\sigma = 1 + O(\alpha'^2)$, so the absence of an oscillator-induced dimension-six term survives ordering by ordering in the same-brane supersymmetric sector. For a quark realization the robust experimental statement is consequently an $O(E^4/M_s^4)$ deformation of hard angular distributions, not necessarily the single function in Eq. (17).

Crossing gives a complementary discriminator. In the same-sign channel the relevant invariants t, u are spacelike and the result is smooth. In an allowed particle–antiparticle channel a Veneziano factor contains a timelike argument s and develops poles at

$$s = nM_s^2, \qquad n = 1, 2, \ldots, \tag{30}$$

whose residues are the Regge polynomials illustrated in Eq. (12). Thus precision angular distortions below threshold and direct Regge production above threshold are not separate hypotheses; they are crossed manifestations of the same analytic amplitude.

Bootstrap interpretation and its obstruction.— The calculation above establishes a limited but nontrivial statement: spatial extension does not force the leading elastic-scattering correction to be a conventional radius term. A Ramond string may be extended on the scale $\sqrt{\alpha'}$ while its tree-level massless scattering agrees with a point-fermion gauge theory through dimension six. This is precisely the property required by a bootstrap picture in which an apparently elementary fermion is re-described by endpoint constituents plus a string degree of freedom.

Chan–Paton algebra makes the constituent bookkeeping suggestive. An oriented endpoint pair gives $N \otimes \bar{N}$, while same-orientation pairs give $N \otimes N$ (or its orientifold projection). For $SU(5)$ this contains

$$5 \otimes \bar{5} = 1 \oplus 24, \qquad 5 \otimes 5 = 10 \oplus 15, \tag{31}$$

mirroring meson-like and diquark-like label sectors. For QCD itself Eq. (1) allows an antisymmetric pair of antiquarks to carry the color of a quark, the same algebraic observation underlying quark–diquark hadronic supersymmetry and modern light-front versions of it [10]. One may therefore formulate a conjectural bootstrap identification

$$q \quad \longleftrightarrow \quad [\bar{q}\bar{q}]_{3_c} \otimes R_{\text{string}}, \tag{32}$$

where the Ramond string supplies the half-integer statistics and whatever additional topological quantum number is needed. In particular $B([\bar{q}\bar{q}]) = -2/3$ differs from $B(q) = 1/3$ by one unit, so a literal realization must assign nontrivial baryon topology to the string/junction sector rather than merely relabel two endpoints.

There is, however, a decisive limitation. Standard RNS supersymmetry does not permit a hadron-sized NS string and an independently point-sized Ramond partner: both sectors use the same X^μ oscillators and the same tension $T = (2\pi\alpha')^{-1}$. A pion-like Regge slope therefore fixes the same string scale for its Ramond sector. Ordinary light-hadron trajectories have slopes near $\alpha'_{\text{had}} \simeq 0.9\ \text{GeV}^{-2}$, corresponding parametrically to $M_s \sim 1.05$ GeV. If Eq. (14) with that same α' described hard quark scattering literally, the contradiction would be immediate. At 90° and only $\sqrt{s} = 10$ GeV one has $z \sim 90$ and Eq. (23) gives a suppression of order 2^{-180} in the cross section, apart from the power prefactor. Instead, perturbative-QCD jet and deep-inelastic data exhibit hard pointlike parton scattering to vastly larger momentum transfers; dijet angular distributions are in fact used to constrain quark contact interactions [11]. Thus a common-tension, hadron-scale open string cannot simply be the observed quark over the experimentally tested hard-scattering domain.

This negative result is as useful as the positive low-energy one. The calculation says that such a fermionic string can *scatter pointlike below its common string scale*; it does not allow the common scale itself to be simultaneously hadronic for the boson and arbitrarily high for the fermion. Consequently Eq. (32) can survive only if "quark = anti-diquark plus string" is a dual/operator identification whose hard amplitudes are not those of the naive common-tension disk, or if an additional mechanism separates the QCD flux-tube scale from the scale controlling the Ramond scattering form factor. That required separation is an experimentally falsifiable dynamical ingredient, not a matter of terminology.

Conclusion.— Four-Ramond disk scattering gives a concrete answer to the operational question "how pointlike can a fermionic string look?" In the simplest same-stack sector, extremely pointlike: after the Chan–Paton projection to a Møller-like charged channel, the exact disk amplitude is the gauge-theory amplitude times Eq. (16), and the leading deviation is $O(E^4/M_s^4)$. The result supplies a precise scattering mechanism for an old bootstrap intuition while simultaneously exposing its main obstruction: pointlike low-energy scattering and microscopic size are not synonymous, but NS and R strings of one theory cannot be assigned independent tensions. Any quark-as-anti-diquark-plus-fermionic-string model must therefore explain the common string scale, baryon topology, and the absence (or interpretation) of the associated Regge structure; those are experimentally meaningful requirements rather than semantic questions about compositeness.

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